🏦 Credit Risk Prediction System

Machine Learning Pipeline for Loan Default Prediction

0.786	102	307K	7
ROC-AUC Score	Features	Training Samples	API Endpoints

Demo Documentation | March 2026

Built by 7afnawi for Hefny

📋 Project Overview

This system predicts the probability of a client defaulting on a loan using machine learning. It includes:

- Complete ML pipeline (Data Engineering → Feature Engineering → Training → Evaluation)
- REST API with FastAPI (7 endpoints)
- Interactive Streamlit Dashboard
- Docker containerization for deployment
- SHAP-based model explainability

✅ **Best Model:** CatBoost with Optuna hyperparameter tuning achieving AUC = 0.786

🎯 Dashboard - Prediction Interface

The main prediction interface allows users to input client information and get instant risk assessment.

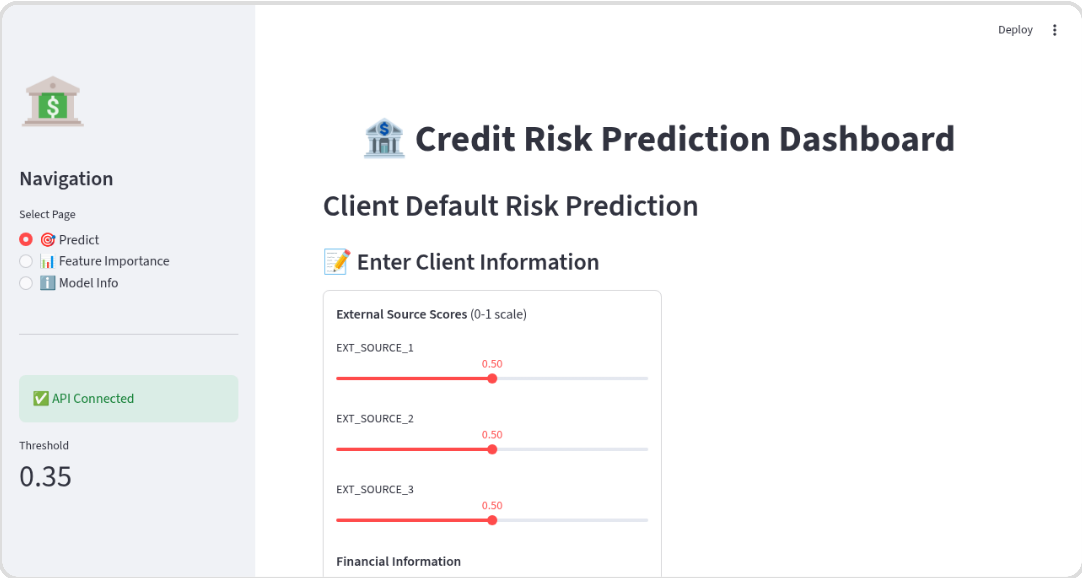

Figure 1: Dashboard home page with client input form

Features:

- **External Source Scores:** Three external credit scores (0-1 scale)
- **Financial Information:** Income, credit amount, annuity
- **Personal Information:** Age, employment years
- **Real-time API Status:** Shows connection status and threshold

📊 Feature Importance Analysis

SHAP-based feature importance shows which features have the most impact on predictions.

Figure 2: Top 20 most important features (SHAP values)

Top 5 Features:

Rank	Feature	Importance
1	FLAG_DOCUMENT_15	6.42
2	REG_REGION_NOT_LIVE_REGION	4.08
3	AMT_REQ_CREDIT_BUREAU_WEEK	2.96
4	INS_PAYMENT_DIFF_MEAN	2.73
5	NAME_EDUCATION_TYPE	2.67

ℹ️ Model Information

Detailed information about the deployed model and its configuration.

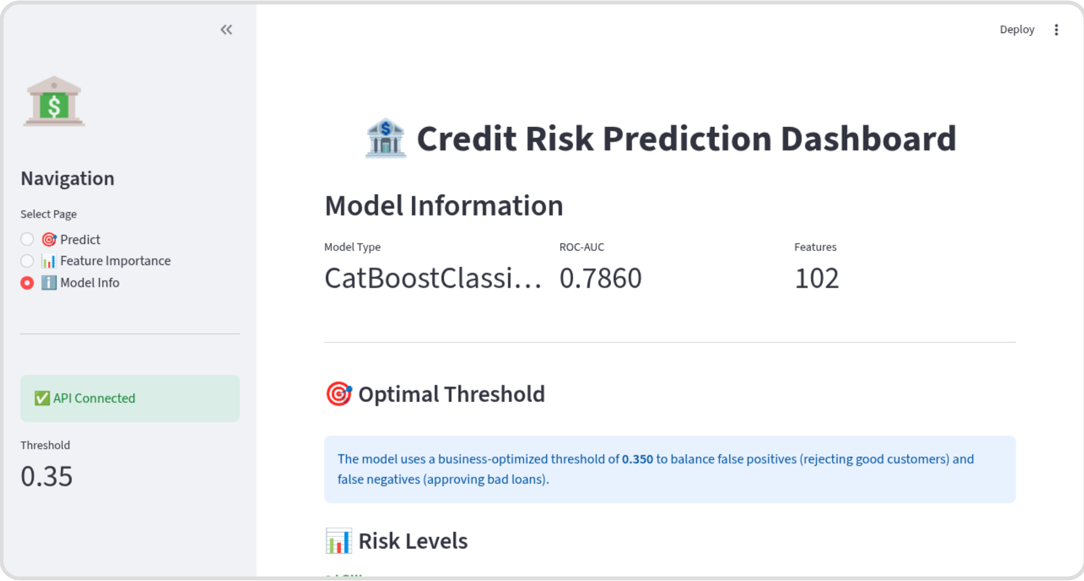

Figure 3: Model information and configuration

Model Specifications:

Parameter	Value
Model Type	CatBoostClassifier
ROC-AUC	0.786
Features	102
Optimal Threshold	0.35
Recall	72%

ℹ️ **Threshold Optimization:** The 0.35 threshold is optimized for business costs, balancing the cost of false negatives

($10,000 for bad loans) against false positives ($500 for rejected good customers).

📡 API Documentation

The system exposes a REST API with 7 endpoints for integration.

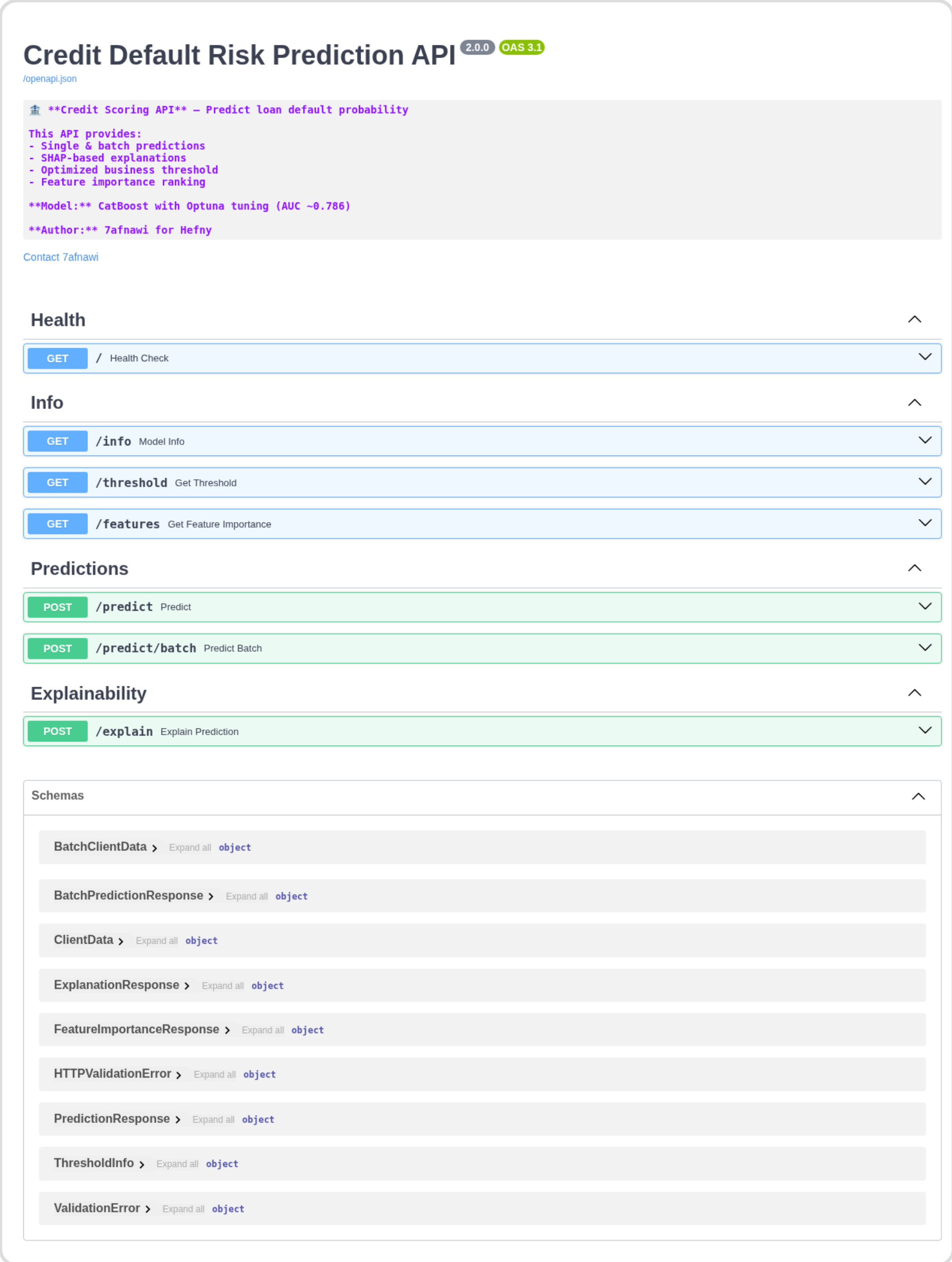

Figure 4: Swagger API documentation

Available Endpoints:

Method	Endpoint	Description
GET	/	Health check
GET	/info	Model information
GET	/threshold	Threshold configuration
GET	/features	Feature importance
POST	/predict	Single prediction
POST	/predict/batch	Batch predictions
POST	/explain	SHAP explanation

Example API Call:

```
curl -X POST http://localhost:8000/predict \ -H
"Content-Type: application/json" \ -d '{ "features":
{ "EXT_SOURCE_1": 0.5, "EXT_SOURCE_2": 0.6,
"AMT_CREDIT": 500000, "AMT_INCOME_TOTAL":
150000 } }'
```

Example Response:


```
{ "client_id": "client_001", "default_probability":
0.232554, "prediction": 0, "risk_level": "LOW-
MEDIUM", "threshold": 0.35, "confidence": "Medium" }
```

🐳 Docker Deployment

The application is fully containerized with Docker Compose.

Running Containers:

Container	Port	Status
credit-risk-api	8000	✅ Healthy
credit-risk-frontend	8501	✅ Healthy

Quick Commands:

```
# Start all services docker-compose up -d # View logs docker-compose logs -f # Stop services docker-compose down
```

🔒 Security Audit Results

Test	Result
SQL Injection	✅ PASS
XSS Protection	✅ PASS
Input Validation	✅ PASS
Container Isolation	✅ PASS
Sensitive Files	✅ PASS

Overall Security Score: 8/10 - Safe for demo and development use.